

PC-Based Equipment Controller

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Presented here is a Windows-based project that can control up to eight electrical devices using a personal computer. Connecting a computer to external devices is becoming essential in our day-to-day life for automation. But to communicate to a device we need a common communication protocol such as a serial COM port, USB or wireless connectivity. Here we have used the serial communication protocol to control the devices.

Many modern personal computers have no serial COM ports but these are still found in industrial machines or scientific instruments. This project was inspired by the construction project, 'Multiple Device Switching Through PC's Parallel Port' published in May 2008 issue of *EFY*. Parallel ports are not in use any

more, nor are they available on PCs and laptops. So we have designed this project for the serial port, which is still being used for communication in some PCs. The author's prototype is shown in Fig. 1.

Circuit and working

The circuit diagram is shown in Fig. 2. It consists of microcontroller AT89C2051 (IC1), MAX232 driver (IC2), eight opto-TRIAC drivers MOC3041 (IC3 through IC10), eight BT136 TRIACs (TRIAC1 through TRIAC8), a DB9 serial COM port connector, a 5V voltage regulator (IC11) and a few other common components.

Power supply. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12 V, 500mA. The transformer output is rectified by full-wave rectifier module BR1, filtered by capacitor C6 and regulated by IC 7805 (IC11). Capacitor C7 bypasses the ripples present in regulated supply. The regulated 5V is used to power the microcontroller,

MAX232 driver, MOC3041 opto-TRIAC drivers and BT136 TRIACs. LED1 indicates presence of power supply.

Microcontroller. The AT89C2051 is a low-voltage, high-performance CMOS 8-bit microcontroller with 2k bytes of Flash programmable and erasable read-only memory. The device is manufactured using Atmel's high-density non-volatile memory and is compatible with the industry-standard MCS-51 instruction set. It has 128 bytes of RAM, 15 I/O lines, two 16-bit timer/counters, a five-vector two-level interrupt architecture, a full duplex serial port, a precision analogue comparator, an on-chip oscillator and clock circuitry. An 11.0592MHz crystal is used as the external clock source. You may use two 33pF capacitors along with the crystal

PARTS LIST

Semiconductors:

- IC1 - AT89C2051 microcontroller
- IC2 - MAX232 driver
- IC3-IC10 - MOC3041 opto-TRIAC driver
- IC11 - 7805, 5V regulator
- BR1 - 1A bridge rectifier module
- TRIAC1-TRIAC8 - BT136 TRIAC
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- R2-R9 - 560-ohm
- R10-R17 - 330-ohm
- R18-R25 - 150-ohm
- R26 - 680-ohm

Capacitors:

- C1 - 2.2 μ F, 16V electrolytic
- C2-C5, C7 - 0.1 μ F ceramic disk
- C6 - 1000 μ F, 35V electrolytic
- C8, C9 - 33pF ceramic disk

Miscellaneous:

- X_{OSC}1 - 11.0592MHz crystal oscillator
- DB9 - 9-pin COM port connector
- CON1-CON9 - 2-pin 5mm terminal connector
- X1 - 230V AC primary to 12V, 500mA secondary transformer
- S1 - Tactile switch

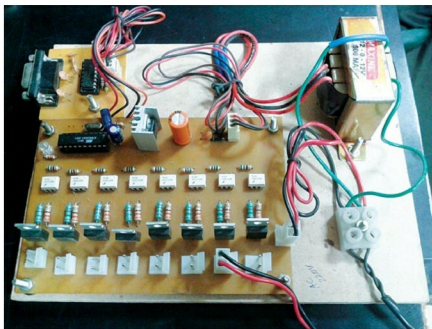


Fig. 1: Author's prototype